

JavaScript Assignment: Rest & Spread Operators

Part A – Multiple Choice Questions (MCQs)

MCQs on Rest Operator

1. What does the **rest operator (...)** do in function parameters?
 - a) Expands an array into elements
 - b) Collects arguments into an array
 - c) Merges two arrays
 - d) Copies objects

 2. Which function parameter declaration is correct using rest?
 - a) `function test(a, b, ...c) {}`
 - b) `function test(...a, b) {}`
 - c) `function test(a, ...b, c) {}`
 - d) `function test(...a, ...b) {}`

 3. Output of:
 1. `function display(...args) {`
 2. `console.log(args.length);`
 3. `}`
 4. `display(10, 20, 30);`
 - a) 2
 - b) 3
 - c) [10, 20, 30]
 - d) Error

 4. In array destructuring:
 1. `const [a, b, ...rest] = [1, 2, 3, 4, 5];`
 2. `console.log(rest);`

Output is:

 - a) 3
 - b) [3, 4, 5]
 - c) [1, 2, 3]
 - d) Error
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MCQs on Spread Operator

5. What does the **spread operator (...)** do?
- a) Collects elements into an array
 - b) Expands elements of array/object
 - c) Deletes object properties
 - d) Declares variables
6. Which is correct to merge arrays `arr1` and `arr2`?
- a) `let result = arr1 + arr2;`
 - b) `let result = [...arr1, ...arr2];`
 - c) `let result = arr1.push(...arr2);`
 - d) `let result = arr1.concat(...arr2);`
7. Output of:
- ```
1. const arr = [1, 2, 3];
2. console.log([...arr, 4, 5]);
```
- a) [1, 2, 3, 4, 5]
  - b) [4, 5, 1, 2, 3]
  - c) [[1, 2, 3], 4, 5]
  - d) Error
8. For objects:
- ```
1. const obj1 = { a: 1, b: 2 };  
2. const obj2 = { b: 3, c: 4 };  
3. const merged = { ...obj1, ...obj2 };  
4. console.log(merged);
```

Output is:

- a) { a: 1, b: 2, c: 4 }
 - b) { a: 1, b: 3, c: 4 }
 - c) { b: 3, c: 4 }
 - d) Error
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Part B – Coding Questions

Rest Operator – Coding

1. Write a function `multiplyAll` using rest that multiplies any number of arguments.
1. `multiplyAll(2, 3, 4); // Output: 24`
 2. Use rest operator in array destructuring to store first two elements separately and remaining in another array.
1. Input: `[10, 20, 30, 40, 50]`
2. Output: `first=10, second=20, others=[30, 40, 50]`
 3. Create a function `showNames` that takes a student's name and unlimited hobbies using rest. Print them in a formatted way.
1. `showNames("Rahul", "Reading", "Cycling", "Coding");`
2. `// Rahul's hobbies are: Reading, Cycling, Coding`
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Spread Operator – Coding

4. Merge two arrays `[1, 2, 3]` and `[4, 5, 6]` using spread and print the result.
5. Copy an object `{ name: "Alice", age: 25 }` into a new object and add a property `country: "India"` using spread.
6. Given:
7. `const numbers = [5, 1, 9, 3];`

Use spread to find the maximum number with `Math.max`.

8. Clone an array `[10, 20, 30]` using spread, and then add new elements `[40, 50]` without changing the original.
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